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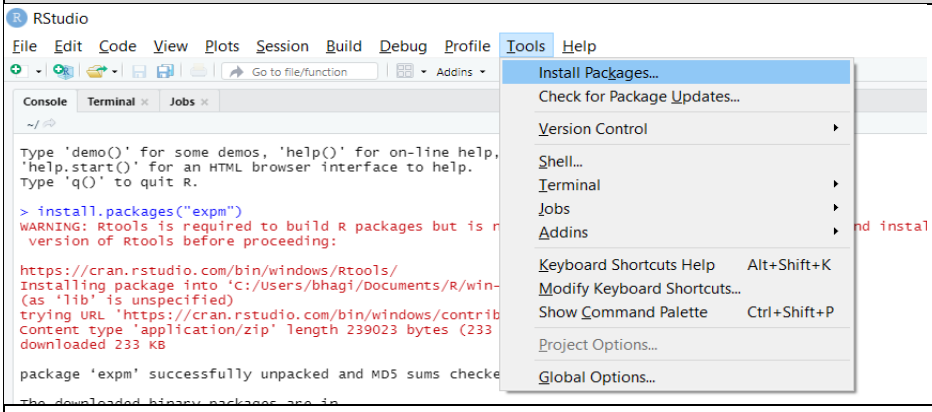
Experiment No.:2

Name of Experiment: Install Packages and calling Installed Packages.

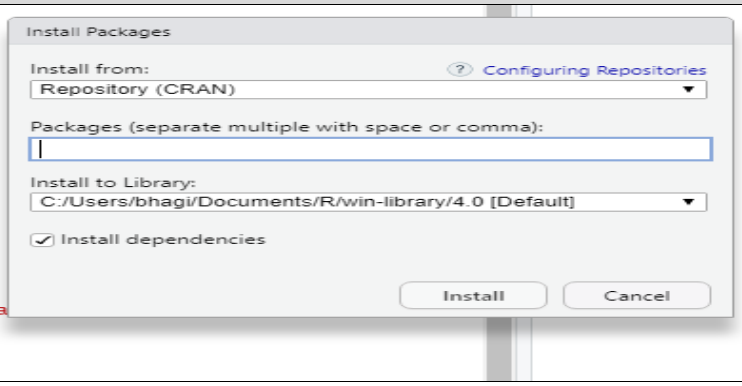
Aim: To Install Packages.

- **R programming** is a very popular language and to work on that we have to install two things, i.e., R and RStudio. R and RStudio works together to create a project on R.

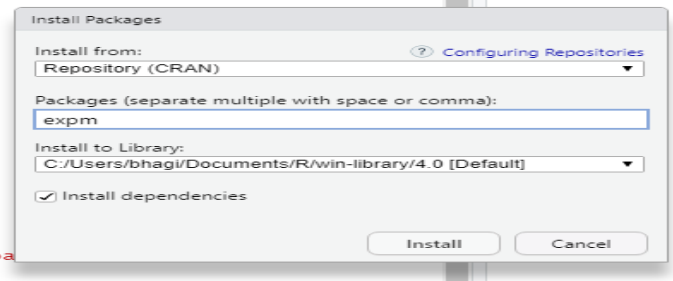
Installing packages in Rstudio.
Method 1: Using application options
1. Open R studio.
2. Select tools



3.After selecting the tools you need to press install packages.



4. Here you need to give the package name you need to install.



Here we used expm. This function computes the exponential of a square matrix.

Method 2: Using command

In this method, simply pass the package to be installed as an argument to install.packages() function

Syntax:

install.packages("package name")

Example:

install.packages("ggplot2")

Loading a package

A package can be loaded once it has been installed using **library()** command.

Syntax;

library("package_name")

Example:

library("ggplot2")

Basic Level Questions

1. What function is used to install a package in R using command-line?
   Answer: install.packages()
2. Which function is used to load an installed package into the current R session?
   Answer: library()
3. What does the package ggplot2 help with in R?
   Answer: Data visualization and graphing.
4. Can you install R packages using the RStudio GUI?
   Answer: Yes, by navigating to Tools > Install Packages**.**
5. What do you need to enter in the package installation dialog box in RStudio?
   Answer: The name of the package to be installed **(**e.g., dplyr, ggplot2).

Moderate Level Questions

1. What is the difference between install.packages("ggplot2") and library("ggplot2")?
   Answer: install.packages() downloads and installs the package, while library() loads it for use in the current session.
2. What might happen if you try to load a package that hasn't been installed?
   Answer: R will throw an error like: *"there is no package called 'package_name'"*.
3. Write R code to install two packages: dplyr and tidyr.
   Answer:

   ```
   install.packages(c("dplyr", "tidyr"))
   ```

4. Why might a package fail to install in R?
   Answer: Possible reasons include: no internet connection, incompatible R version, missing dependencies, or lack of admin rights.
5. How can you verify if a package is already installed in R?
   Answer:

   ```
   "ggplot2" %in% rownames(installed.packages())
   ```

Advanced Level Questions

1. How do you install a package from GitHub using the devtools package?
   Answer:

   ```
   install.packages("devtools")

   library(devtools)

   install_github("username/repo")
   ```